Improving Accuracy & Precision

A Practical Guide for Stock Price Prediction Models

This guide covers the areas that have the greatest impact on model accuracy and precision for financial time-series prediction: target definition, feature engineering, validation methodology, model selection, overfitting control, and realistic benchmarking.

1. Rethink What You're Predicting

Predicting the absolute price for tomorrow is the hardest and least useful target, since it is dominated by today's price (high autocorrelation makes naive models look falsely accurate). Instead:

- **Predict returns** (percentage change) or **log returns**, which are stationary and easier for models to learn from
- **Predict direction** (up/down) as a classification problem — often more useful and more tractable
- **Predict volatility / risk bands** instead of a single point estimate

A model that correctly predicts direction 55–58% of the time *consistently* is considered quite good and potentially tradeable, even if it cannot pinpoint exact future prices.

2. Feature Engineering Is Where Most Gains Come From

Raw OHLCV (open/high/low/close/volume) data alone is a weak signal. Strong feature sets typically include the following categories.

Category	Examples
Technical indicators	RSI, MACD, Bollinger Bands, ATR, moving average crossovers, momentum / rate-of-change
Lagged returns	1-day, 5-day, and 20-day return windows
Volatility measures	Rolling standard deviation of returns
Volume-based features	Volume spikes, on-balance volume
Market-wide context	Broader index movement (e.g., S&P 500), sector performance, VIX
Calendar effects	Day of week, month, proximity to earnings announcements
Sentiment data	News sentiment scores, social media sentiment (often a real source of edge, but requires an NLP pipeline)
Fundamental data	P/E ratios, earnings surprises — useful for longer-horizon predictions

3. Validation Methodology — Critical and Often Done Wrong

Standard random train/test splits leak future information into training and produce wildly optimistic results that fall apart in production. Instead, use:

- **Walk-forward validation** (rolling-window backtesting): train on data up to time T, test on T+1 to T+k, then slide the window forward
- **Never shuffle** time-series data
- Compute scaling/normalization statistics only on training data, then apply them to test data — never "peek ahead"
- Watch for **survivorship bias** when backtesting across a universe of stocks (don't only test on companies that still exist today)

> Validation methodology errors are the single most common reason a model appears highly accurate in testing but performs poorly in live use.

4. Model Selection & Ensembling

No single model architecture is a silver bullet. What tends to work well in practice:

- **Gradient boosted trees** (XGBoost / LightGBM) on engineered tabular features — often outperform deep learning on financial data because they handle noisy, non-stationary data with fewer parameters to overfit
- **LSTM / GRU or Temporal Fusion Transformers** if sufficient data is available and sequential patterns need to be captured directly
- **Ensembling** multiple models (e.g., averaging or stacking XGBoost with an LSTM) often beats either model alone, since each captures different patterns
- **Quantile regression** or models producing prediction intervals (not just point estimates) — useful for confidence-interval visualizations and more honest about uncertainty

5. Reducing Overfitting

- **Regularization:** L1/L2 penalties, dropout for neural networks, max depth / min samples per leaf for tree-based models
- **Cross-validation across multiple stocks and time periods**, not just one stock — a model that only works on one ticker's history is likely overfit to noise
- **Keep the feature set lean:** more features often hurts on noisy financial data unless there is a large amount of training data
- **Early stopping** based on validation loss

6. Realistic Benchmark to Beat

Always compare new models against simple baselines:

- A naive "tomorrow equals today" model
- A simple moving average
- A basic linear regression on lagged features

If a more complex model (e.g., an LSTM) does not meaningfully beat these baselines on out-of-sample, walk-forward tests, the added complexity is not earning its keep.

In published research and industry practice, directional accuracy in the high 50s to low 60s percent range on daily horizons is generally considered solid. Any model claiming 90%+ accuracy on raw price prediction is almost certainly suffering from data leakage or overfitting, and should be re-validated using walk-forward methodology before trusting the result.